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Conference Paper · November 2016

DOI: 10.1109/ICAMSE.2016.7840358

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Investigating elementary students' epistemological beliefs, game preference by applying game-based learning to a history course

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Abstract

In this study, a game integrated historical events were designed for improving and promoting students' motivation and understanding for history education. For this reason, the epistemological beliefs were incorporated in the model to explore students' digital games preference by applying technology acceptance model (TAM). A total of 112 fifth gradestudents in an elementary school were participated in a four-week digital history game learning. A partial least squares (PLS) technique applied to verify that intention to use digital games was directly influenced by students' game preferences and epistemological beliefs. Therefore, educators should recognize and enhance students' epistemological beliefs when implementing the game-based learning, and increase students' game learning effectiveness.

Keyword: game-based learning, ease of use, usefulness, epistemological beliefs, game preferences, behavioral intention, technology acceptance model

Introduction

The traditional history learning in Taiwan is that students listen to a lecture delivered by the instructors based on direct instructional approaches. Students are used to adopt externally directed approaches (i.e., teacher-centered rather than student-centered learning), thus students became passively learning and lack the abilities to analyze and integrate related information across different sources. Such demanding processes are probably to some extent guided by people's understanding of what knowledge and knowing is, which referred as epistemological beliefs.

The epistemological beliefs for students' learning performance have come into prominence in educational literature ([1], [2]). Bromme et al. (2010) predicted and interpreted students' epistemic beliefs affected the learning content with students' learning engagement [3]. One of the few empirical studies of digital history game learning was conducted by Moshirnia & Israel (2010), found that retention effects and attention were better in gameplay groups than traditional groups [4]. In history education, game-based learning can promote conceptual understanding over simple declarative knowledge to enhance historical thinking rather than recalling historical facts [5].

According to Ossianilsson (2012), technology acceptance model and computer assisted learning accelerates the change of academic learning, but more emphasis should be put on knowledge structure constructions [6]. However, the relationships between students' epistemological beliefs and their learning processes in game-based learning have received surprisingly little scholarly attention. Therefore, there are

needs for researches on how epistemological beliefs influence game preferences and motivate students to active participate in constructing learning knowledge.

This study has attempted to fill this gap by applying incorporating epistemological beliefs into our research model. The aim of this study was to extend the TAM to explore the factors related to game-based learning user behaviors. Specifically, this work proposed additional variables, for example, epistemological beliefs, enhanced our understandings of game user behaviors. The importance of epistemological factors can explain with reference to existing literature on influences in the field of educational psychology.

Literature review

Technology acceptance

In the past two decades, the TAM has become recognized as one of the most widely used and empirically tested model in information technology [9]. Legris (2003) argued that the TAM findings are not always consistent [10]. One of the major problems is the inability of the model to account for individual, contextual, and task characteristics [11]. Researchers tried to solve this problem by extending the model with potential factors suggested by other theories. These extending models not only contributed to a deeper understanding of technology acceptance, but also they help to explain larger proportions of variance.

Digital game acceptance

As the body of research points out the potential educational value of games grows, the interests for including digital games learning in educational contexts is increasing in popularity [12]. Generally speaking, digital games learning refers to the use of games to enhance the learning experience, while maintaining a balance between contents and games [13]. Researchers have applied the TAM model in connection with digital games learning acceptance [11]. Park et al. (2013) found that perceived ease of use and usefulness were the major factors for users' acceptance of mobile games [14]. Bourgonjon et al. (2010) further studied that the students' preference for using video games and believed that preference is affected directly by usefulness, ease of use, learning opportunities, and personal experience [11]. Even though there are several studies employed the TAM in game-based learning.

Epistemological beliefs

In generally, epistemological beliefs concern individual's conceptions about the nature of knowledge and knowing. Many researchers suggested that the epistemological beliefs of learners are positively associated with academic performance. Historically, studies on epistemological beliefs have been started with

unidimensional approach. In this approach, researchers have viewed personal epistemologies as developmental stages. Based on Perry (1970) longitudinal research, suggesting that learners' epistemological beliefs develop through a set of stages: from "naïve" beliefs as they entering the university to more "sophisticated" beliefs as they advanced through their education [15].

Research model and hypotheses

The research framework is constructed according to the relevant literature, as shown in Figure 1. Based on the theoretical background and 9 postulated hypotheses. Variables include students' perceived ease of use, perceived usefulness, epistemological beliefs, preference, and intention to use the system.

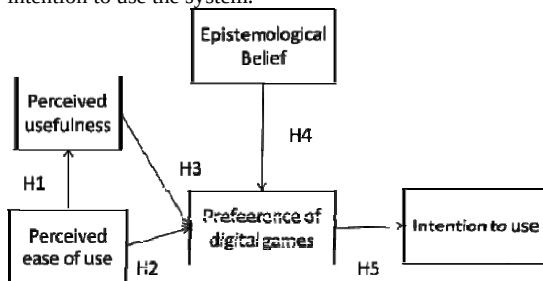


Figure 1. Research model

Perceived Ease of use

Perceived ease of use is a prime determinant for behavioral intention in TAM. It is defined as the degree to which a person believes that using a particular system would be free of effort and has a direct, positive influence on perceived usefulness. Therefore, the TAM reasons that people will perceive a certain type of technology as more useful when it is easy to use [11]. It can be hypothesized that people will perceive a certain type of technology as more useful when it is easy to use. In recent study, Bolliger et al. (2015) investigated Japanese university students' perception of games in English learning and found that perceived ease of use and usefulness positively influenced preference for video games [16].

H1: Perceived ease of use (EOU) positively affects usefulness

H2: Perceived ease of use (EOU) positively affect preference for digital games.

Perceived usefulness

Perceived usefulness is another key predictor of intention to use a technology in TAM. Davis defined usefulness as "the degree to which a person believes that using a particular system would enhance his/her job performance (Davis, 1898). The use of the word "performance" in this definition suggests that it should be focused on results. From the educational perspective, the learning process itself can be equally important as the outcome. Therefore, we replaced the attitude by preference toward games.

H3: Usefulness positively affects preference for digital games.

Epistemological beliefs and preference

Epistemological beliefs have been the subject of study in relation to learning processed [2] and categorized into naïve beliefs and sophisticated beliefs. Naïve beliefs could be characterized as straightforward with regard to the nature of knowledge. In contrast, a person with more sophisticated beliefs perceived knowledge as dynamic, interrelated and more constructed rather than found. In this study, we proposed that a person with more sophisticated beliefs would involve in the game-based learning environment and resulted in intention to use.

H4: Students' with sophisticated beliefs of knowledge is positively affects preference for digital games.

Preference for digital games

The acceptance of game-based learning depends on the preference digital games. It is derived from Hsu & Lu (2007), who described preference as a measure of the degree of user' positive feeling about participating. In addition, Bourgonjon et al. (2010) added an intended behavior component that refers to the approval of game-based learning in order to examine students' acceptance of digital games [11]. Studies should considered the preferences of game-based learning that may contribute to understand the digital games adoption.

H5: students' preference of digital games affect intention to use digital games learning.

Design of experiment

In this study we applied an extended TAM model that takes a multimodal analytical approach to examine how computer games shape students' exploration during a history learning activity. The data for this study was collected using a survey comprising questions on demographics and item scales for measuring the different variables in the research model. Structure equation modeling (SEM) was used to analyze the different predictors for students' preference of digital games and the relationships between these predictors.



Figure2. Digital game setting

Procedure

The study was conducted in four weeks of elementary history course in Taiwan. Students participated in the experiment need to complete the assignments instructed by games (Figure 2). Therefore, students need to search for the clues and ask the key person for assistance in the games.

The assignments including multiple view points and resources for open-ended exploration. At end of the history game learning experiment, students took the instrument.

Data Analysis and Results

The partial least squares (PLS) technique was used for data analysis. PLS is a second generation regression model that confirmatory factor analysis with linear regression, and makes it possible to run the measurement and structural models simultaneously. We used PLS because our sample size was relatively small. The statistical software Smart PLS 2.0 was used to access the measurement and structural models for our research. Table 1 provides an overview of the path coefficients. Both Perceived ease of use and usefulness appear to be significant predictors for students' preference for digital games. Perceived ease of use also influence the learners' usefulness in digital games use. Additionally, the preference of digital games has a positive effect on intention to use. Summarizing, all hypotheses constructing the original TAM are confirmed in this study. The results of the analysis also show that the new constructs, sophisticated beliefs are positively affect preference.

Table 1. Testing the hypotheses

Hypothesis	Effects	Coefficient	S.E.
H1	Ease of use → usefulness	0.34*	0.225
H2	Ease of use → preference	0.21*	0.020
H3	Usefulness → preference	0.32*	0.025
H4	Sophisticated beliefs → preference	0.31*	0.036
H5	Preference → intention	0.38*	0.132

* $p < .05$

Discussion

Our study presented and validated the extended TAM model to assist in understanding the factors that contributing to preference of digital games. With empirical analysis, several implications were obtained. First, perceived ease of use and usefulness appeared to be important variables in the game-based learning. Especially, usefulness was found to have the most significant influence on game preference. These results are supported by previous studies [11].

Second, according to previous researches that the role of game features fostering students' learning and motivation (Wouters & Van Oostendorp, 2013) were also supported in our study. In addition, Cózar-Gutiérrez & Sáez-López (2016) showed that the game-based learning in history could involve a higher level of learning activities and engagements [17]. Thus, the learner would be interested in exploring ways in which increased the understandings of history learning.

Finally, the epistemological beliefs were chosen for incorporating into our conceptual model for both empirical and theoretical reasons. The traditional education is teacher-centered approach and makes students learn passively. In contract, the student-centered learning (game-based), can better attract students than traditional teaching methods. Pedagogical principles alone cannot constitute an interesting and attractive game that motivates people to play. Digital games provide opportunities for

authentic, reflective, and critical thinking practices that are not only pertinent, but essential for successful learning in the 21st century.

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